Function Overloading:

As we know C++ is object-oriented programming language. Means it provide us features to implement the principle of oops.

And **Polymorphism** is one of the principle of oops.

Polymorphism:
To take more than one form

or

polymorphism allows us to write more flexible and reusable code by enabling objects of different types to be treated as object of common base.

There are two main types of polymorphism in C++

- Compile-time polymorphism (static polymorphism) {Overloading}
  - Function overloading
  - Operator overloading

  → Achieved through Overloading
- Run-time polymorphism (dynamic polymorphism) {Overriding}
  - Virtual function
  - inheritance function → member function of a base class that is available in the derived class through inheritance.

And Run-time polymorphism is achieved through overriding

Function Overloading:

- Function overloading is a way of implementing polymorphism in C++
- Overloading refers to the use of the same thing for different purposes.

* Use of same function name to create functions that perform a variety of different tasks. is known as function overloading in OOP.

Example :

```
int add (int a, int b) ;                    // prototype 1
int add (int a, int b, int c);              // prototype 2

double add (double x, double y);            // prototype 3
double add (int p, double q);               // prototype 4
double add (double p, int q);               // prototype 5

cout << add(5,10);          uses prototype 1
cout << add (15, 10.0);     uses prototype 4
cout << add (12.5, 7.5);    uses prototype 3
cout << add (5,10,15);      uses prototype 2
cout << add (0.75,5);       uses prototype 5
```

How function Overloading is Resolved?

A function call first matches the prototype having the same number and type of arguments and then calls the appropriate function for execution.

The function selection involves the following steps:

Step-1 The compiler first tries to find an exact match in which the types of actual arguments are the same and use that function.

Step-2 If an exact match is not found, the compiler uses the integral promotions to the actual arguments such as

char to int
float to double